

Cybersecurity Concepts Lab for Teachers

Trust your hackles. Pause · Check · Tell.

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 12: Spot the Fake (Deepfakes, Voice Clones and Image Provenance, Unplugged)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE	
Big idea	We can no longer trust our eyes and ears alone. Instead of trying to <i>spot</i> fakes by their flaws, we build verification habits : find the source, find the original, find other reliable confirmation, and use a second channel for urgent requests.
Grade bands	K-2 (real or pretend?), 3-5 (caption flip + trail detectives), 6-8 (full Fake Lab), 9-12 (full + provenance technology and the "liar's dividend")
CSTA-style concepts	Impacts of Computing (ethics, trust, privacy), Data and Analysis (evidence), Networks and the Internet (cybersecurity) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed case envelopes, index cards, pencils. No devices; no real fake images or audio are used or created.
Teachers do it first	Yes. Part A about 25 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Ask a neighbor: "Someone sends you a shocking photo. Name two things you would check before sharing it, neither of which is how it looks." (Source? Original? Another report? A second channel?)

01 The big idea

ARTIFICIAL INTELLIGENCE CAN NOW create photos, videos and voices that look and sound convincing (often called **synthetic media**, or **deepfakes** when they imitate real people). But the bigger, older problem is simpler: **real media with false captions** ("cheap fakes"), old images re-shared as new events, and edited clips. The skill we need for all of them is the same: **checking where something came from, not judging how it looks**.

A second big idea, which many adults miss: the opposite mistake. When fakes exist, people can wrongly dismiss **real** evidence as fake (sometimes called the "**liar's dividend**"). Good defenders are not cynical; they are **careful**.

02 What a teacher should know (plain language)

2.1 THE HONEST TRUTH ABOUT "SPOTTING THE GLITCHES"

EARLY ADVICE SUCH AS "look at the hands, the teeth, the blinking" had some value, but it has a short shelf life. Models improve quickly. Studies have found that people are often close to chance at telling high-quality AI faces from real ones, and sometimes rate AI faces as *more* trustworthy or more real [VERIFIED 2026-10-05: Nightingale and Farid, PNAS, February 2022, Crossref abstract: AI-synthesized faces "indistinguishable and more trustworthy than real faces"; the second cited study (Miller et al. 2023) was not checked]. Automatic "AI detector" tools are also imperfect, with both false positives and false negatives [VERIFY current research]. So:

- **Do** mention visual and audio oddities as *clues to look into*, never as proof.
- **Do not** teach "if it looks perfect it's real" or "if it looks odd it's fake".
- **Do** teach **process**: source, original, corroboration, second channel.

2.2 THE KINDS OF FAKES (A QUICK MAP)

TYPE	WHAT IT IS	TYPICAL MISUSE
Cheap fake / miscaption	Real photo or video with false context or small edits (cropped, slowed, relabeled)	Fake news, rumors
Image generation	A picture created entirely by AI	Fake events, fake "evidence", fake profile pictures
Face-swap / video deepfake	A person's face or body made to appear in a video	Fake endorsements, scams, harassment
Voice cloning	AI imitating someone's voice from a short sample [the FTC and FBI both say a short audio clip is enough: VERIFIED 2026-10-05: consumer.ftc.gov, ic3.gov PSA241203; how little varies by tool]	"Relative in trouble" calls, fake bosses, voicemails
Non-consensual intimate deepfakes	Fake sexual images made of real people, including minors	A serious harm; illegal in many places (see Safety)

2.3 VERIFICATION HABITS ("THE FIVE CHECKS")

Built on the SIFT method from digital literacy researcher Mike Caulfield (Stop, Investigate the source, Find better coverage, Trace claims, quotes and media to the original context) [VERIFIED 2026-10-05: hapgood.us, "SIFT (The Four Moves)", 2019]. In kid language:

1. **Stop.** Pause before you react or share, especially if it makes you angry, scared or excited. (Strong feelings are a red flag, just like in Demo 3.)
2. **Source.** Who posted it? Is it the original source? Do I know them?
3. **Another source.** Do other trustworthy sources report it? Searching *outside* the post helps. ("Lateral reading": leave the page to check the page.)
4. **Original.** Where did the picture or clip first appear? (Reverse image search tools can help adults and older students find earlier appearances [VERIFY tool policies].)
5. **Second channel.** For urgent requests from "someone you know", check by another route: call back on a known number, use the family code word (Demo 8), ask in person.

2.4 PROVENANCE TECHNOLOGY (WHAT IS COMING)

- **Content Credentials / C2PA.** An industry standard that lets cameras, editing software and platforms attach **signed information** about where media came from and what edits were made, like a tamper-evident label (the Coalition for Content Provenance and Authenticity) [standard exists, VERIFIED 2026-10-05: c2pa.org; current adoption not checked].
- **Watermarks.** Some AI systems embed hard-to-see signals marking content as AI-made (for example, Google's SynthID, described by Google DeepMind as a tool to watermark and identify AI-generated content)

[VERIFIED 2026-10-05: deepmind.google]. They help for content from participating tools but do not cover everything.

- **Limits.** Labels can be stripped; missing labels do not prove something is real or fake; present labels say what the *signer claims*. It is another layer, not a verdict. Teach "provenance helps; trace and corroborate".

2.5 VOICE CLONES AND THE CODE WORD

Voice clone scams often mimic a family member in distress, urging secrecy and quick payment. The FBI has advised families to consider a **secret word or phrase** to verify identity in emergencies [VERIFIED 2026-10-05: FBI IC3 PSA241203: "Create a secret word or phrase with your family to verify their identity"]. Combine it with the **callback rule** (Demo 8): hang up, look up, call back on a known number. Teach: *a familiar voice is no longer proof*.

2.6 HARMS AND THE SCHOOL CONTEXT (READ CAREFULLY)

- **Non-consensual intimate deepfakes** (including those made from a student's ordinary photos) are a serious harm affecting students. Creating, possessing or sharing sexual images of a minor, even AI-generated ones, may be a crime. Laws are changing quickly (for example, the US TAKE IT DOWN Act of 2025 and state laws) [VERIFY current federal and state law]. Schools need a clear procedure.
- **Targets are never at fault.** Responses: do not forward, do not view more than needed, preserve evidence only as per school policy, **report to a trusted adult and the school's designated person**, use removal services such as NCMEC's Take It Down for images of minors [VERIFIED 2026-10-05: takeitdown.ncmec.org]. Follow the disclosure steps in your band's Teacher Guide and the Index box.
- **Never create fake images of real students or staff in class**, even as a "joke" or "demo".

2.7 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"I can always tell a fake."	Often not, and it gets harder over time.
"AI detectors will tell me."	They are unreliable.
"If it's on video it's true."	Video can be fake or out of context.
"If it looks fake it must be harmless."	Even crude fakes can be used to harass or scam.
"If I doubt everything I'm safe."	Total distrust makes people miss real warnings. Verify, don't dismiss.

03 Materials (zero to low cost)

- **Case Envelopes** (6): each holds a claim card, a "media" card (a description or a hand-drawn cartoon, not a real image), and three or four evidence cards (source trail, other reports, official statement).
- **Caption Flip cards**: 4 drawings (a crowd, a flooded street, a lone person, a fire truck) and 8 captions.
- **Provenance Chain** cards: 6 cards for the relay.
- **Rating tokens**: Verified (green), False (red), Real but Miscaptioned (yellow), Not enough info (gray).
- A timer, chart paper, student handout.
- **Family code word card** (a blank card for a fake code word, as in Demo 8).

04 The activities

4.1 WARM-UP: THE TELEPHONE PROVENANCE RELAY (5 MINUTES)

SEAT SIX STUDENTS in a line. Student 1 gets a card describing a simple event (for example, "A blue kite got stuck in the oak tree near the library at 3 p.m. and a firefighter got it down."). Student 1 whispers it to Student 2, and so on. Student 6 writes what they heard. Compare with the original. The details always change. Then: "Every share can change a story. To know what really happened, find the original. That's provenance."

4.2 CAPTION FLIP (ALL BANDS)

Show the same cartoon drawing (for example, a crowd in a square) with two different captions: "Celebration after the school's win" and "Protest against lunch changes". Ask: "Which is true? What would you need to know?" (Who took it? When? Where?) Students create their own caption flips on paper to see how context changes meaning. **Rule:** use drawings only, no real photos of people.

4.3 THE FAKE LAB: CASE ENVELOPES (GRADES 5-12)

Print only the case text below (claim, media, evidence) for the envelopes. The ratings and reasoning are on the teacher answer page. **Case 5 mentions a relative and secrecy; make it optional and let any student pass.**

Teams of 3-4 become **Verification Teams** and get one envelope each (rotate). Each team applies the five checks and uses a token to rate the case, writing a short reason.

Case 1: The giant pumpkin. Claim: "A 2,000-pound pumpkin won at the Greenfield Fair." Evidence: first posted by the Fair's official account; the

local newspaper confirmed with a quote from the judge; many attendees posted.

Case 2: The flooded hallway. Claim: "Greenfield School flooded today! School closed!" Media: a photo of a flooded hallway. Evidence: an image-search card shows the same picture from three years ago in another town; the school's official alert system says nothing.

Case 3: The principal's voice message. Claim: a voicemail in "Principal Reyes's" voice says school is closed Friday and to reply with your parent's email and phone. Evidence: the school's official site and alert system do not mention closures; the number is unknown; the school office says they didn't send it.

Case 4: The star skater's offer. Claim: a video of a famous skater, "Jay Moreno" (fictional), says to join a website to double your money. Evidence: his official account has no such video; a fact-check card says it is a fake; the site is unknown.

Case 5: The panic call. Claim: "Hi, it's me, my phone died, I'm in trouble, I need money right now, don't tell Mom." Evidence: the voice sounds like a relative; a family code word is requested and not given; calling the relative's known number reaches them safe at home.

Case 6: The unexpected owl. Claim: "A rare snowy owl landed on the town hall roof!" A student says "It's AI, owls don't do that." Evidence: three local news sites, a wildlife agency post, and several photographers from different angles.

After each rotation, the teams present: "We rated it ___ because ___." The teacher asks: "Which check gave you the answer?"

4.4 THE SECOND CHANNEL ROLE-PLAY (GRADES 3-12)

Pairs practice a short script (Demo 8 links): **Caller card** (read as printed): "I'm your cousin; my phone broke; I need you to send a gift card; don't tell anyone." **Receiver:** "Hold on, what's our code word?" / "I'll call you back on your number." Switch roles. All names are fictional. No improvisation of persuasion tactics.

4.5 REAL OR PRETEND? (K-2)

Show 6 cards: a drawing of a flying pig, a photo-style card of a classroom, a talking toaster, a person with three arms, a duck at a pond, a cat riding a skateboard. Ask "Real or pretend? How do you know? Who can we ask?" Introduce: "Some pictures are made by computers to look real, even when they're pretend. When a picture or voice asks you for something or says something scary, **ask a grown-up.**"

05 Part A: Teachers do it first (about 25 minutes)

- 1. Cold ratings (8 min).** Without the answers, rate Cases 1-6 with the five checks. Where did you want to trust your gut?
- 2. Compare (5 min).** Case 6 (real but surprising) tests over-skepticism; Case 2 tests whether you checked context.
- 3. Code word and callback (6 min).** Practice the Second Channel script in pairs. How does it feel to say no to a voice that sounds like someone you love?
- 4. Plan (6 min).** How will your class handle sensitive content (non-consensual deepfakes)? Review the school's policy and reporting route, and decide what you will say if a student discloses.

06 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: "Which of these is real?" with the K-2 style cards.
2-5	Telephone Provenance Relay (4 students).
5-10	Caption Flip and discussion.
10-13	One case (Case 2 or Case 5) done as a class using the five checks.
13-15	Exit slip: "Before I share, I ____."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and honest truth: we can't always spot fakes.
5-10	Telephone Provenance Relay.
10-15	Caption Flip.
15-33	The Fake Lab: three case rotations with presentations.
33-40	Second Channel role-play and family code word card.
40-45	Exit ticket and handout.

07 Teacher script

Hook (2 minutes). "Two truths and a fake. I'm going to describe three things: a sandwich shaped like a dinosaur, a dog who can drive a bus, and a photo of a bridge. One is made up. Hard to tell, right? In the real world it's tougher: pictures, videos and even voices can be made by computers. Today we become verification detectives. And here's a secret: the best detectives don't mostly look at the picture. They look at where it came from."

On the glitches. "You may hear 'look at the hands'. That helps sometimes. But computers keep getting better. Don't rely on it. Rely on the five checks."

Case debrief. "Which check solved it? Source? Another source? The original? The second channel? Case 6 is the opposite problem: the owl was real. If you say 'everything's fake', you stop being a detective and become a cynic."

Close. "If a voice, a video or a picture ever makes you feel panic, hurry or a secret: pause, check by another route, and tell. If someone ever makes a fake picture of you or a friend, or sends one around, it is never the target's fault. Tell a trusted adult, and don't forward it."

08 Kid-friendly analogies by band

BAND ANALOGY AND ACTIVITY

K-2	Real or pretend? Drawings of silly and real things. "Pretend pictures can look real. Ask a grown-up."
3-5	The telephone game and caption flip. Pass a message, notice what changes; swap captions on the same drawing. "Find the first teller."
6-8	Verification Team. Five checks on Case Envelopes. Discuss how feelings are used.
9-12	Provenance and trust at scale. Content Credentials, watermarking limits, detector unreliability, the liar's dividend, law and platform policy. Debate: should AI images always be labeled?

09 Discussion questions

1. Why can't we rely on spotting glitches? Why is it harmful to say "everything online is fake"?
2. Which check would have solved the most cases? Which was hardest?
3. What makes a good second-channel check for a message from a friend?
4. (9-12) What are the strengths and limits of labels like Content Credentials? Who decides what gets labeled?

10 Level-up challenge

- **Verification Poster** for a younger class: five checks, a second-channel example and a "who to tell" line.
- **Design a Case Envelope.** Teams write a new *fictional* case with evidence that supports a different rating; another team rates it.
- **School policy (9-12).** Draft a one-page policy on AI-generated images of students that protects targets, discourages sharing and outlines consequences [VERIFY with counsel and administrators before any real adoption].

11 Safety and ethics notes (defense only)

- **Do not create deepfakes or AI-generated images of real people in class.** Do not show examples of non-consensual or sexual synthetic images. Do not use "nudify" or face-swap apps. Do not use any real student's or teacher's photo or voice.
- **Use fictional cases and drawings only.** No real harmful images, audio or video.
- **No "how to make a convincing fake" instruction.** Teach to verify, not to fabricate.
- **Sensitive content.** Some students may have been targeted or know someone who has. Keep the tone calm and supportive, never blaming. Offer a private route to an adult. Do not ask students to share examples.
- **Disclosures.** If a student discloses a synthetic or real sexual image, threats or sextortion, **do not view, copy, forward or ask for the images.** Follow your Teacher Guide's disclosure procedure, involve the designated school contact, and report as required (see the Index box and the "If a student discloses" section of your band's guide) [VERIFY local law and district policy].
- **Take care with "AI detector" tools.** Do not use them on students' work to accuse them; they are unreliable [general guidance: detector tools produce false positives and false negatives; no single primary study was checked in Round 3B].

12 Assessment idea

- **Exit ticket.** "Name two of the five checks and a time you would use the second channel." Rubric: 3 = two checks and a correct scenario; 2 = one check; 1 = "look for glitches".
- **Reflection.** "Name one thing that makes you want to share a post immediately. What can you do instead?"

13 Sources and [VERIFY] list

- Nightingale and Farid, "AI-synthesized faces are indistinguishable from real faces and more trustworthy," PNAS (2022) [VERIFIED 2026-10-05: Crossref record].
- Mike Caulfield, SIFT [VERIFIED 2026-10-05: hapgood.us] and "Web Literacy for Student Fact-Checkers" [not checked].
- C2PA (Coalition for Content Provenance and Authenticity) specification and Content Credentials [VERIFIED 2026-10-05: c2pa.org].
- Google SynthID and other watermarking initiatives [SynthID VERIFIED 2026-10-05: deepmind.google].
- FBI Public Service Announcements on AI-enabled scams and voice cloning [PSA241203 (December 2024) VERIFIED 2026-10-05: ic3.gov].
- FTC consumer alerts on voice cloning [VERIFIED 2026-10-05: consumer.ftc.gov, March 2023].
- US TAKE IT DOWN Act (2025) and NCMEC "Take It Down" service [VERIFIED 2026-10-05: takeitdown.ncmec.org].
- Chesney and Citron on the "liar's dividend" (2019) [not checked].
- Reported 2024 case of a deepfake video call used to steal funds from a company in Hong Kong [widely reported; not verified here, so check a reliable news or police source before citing; the FBI PSA of December 2024 warns generally about real-time video-chat impersonation, VERIFIED 2026-10-05].

Student Handout: Spot the Fake

Name: _____

The big idea

WE CAN'T ALWAYS TELL a fake by how it looks. We check **where it came from**.

The five checks

1. **Stop:** pause before you react or share.
2. **Source:** who posted it? Is it the original?
3. **Another source:** do trustworthy places report it too?
4. **Original:** where did it first appear?
5. **Second channel:** check by calling, or asking in person (use a code word for emergencies).

Caption Flip

SAME DRAWING, TWO CAPTIONS: "Win celebration" / "Protest". What would you need to know to decide? _____

Rate it

Write a rating: **Verified / False / Real but miscaptioned / Not enough info.**

1. A photo of a flooded hallway that is really from three years ago in another town: _____
2. A voicemail in the principal's voice asking for your parents' phone number, not mentioned by the school's official site: _____
3. A surprising owl photo confirmed by three local news sites: _____



My second-channel sentence

"Hold on. What's our code word?" or "I'll call you back on the number I already have."

Strong feelings test

IF A POST MAKES me feel **angry, scared or in a hurry**, I will _____ before sharing.

Remember

- Pictures, videos and voices can be faked, but real things exist too: don't say everything is fake.
- If someone makes or shares a fake picture of you or someone else, it is **never the target's fault**. Don't forward it. **Tell a trusted adult**.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 12 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Handout: 1 Real but miscaptioned; 2 False (verify with the office; do not reply with personal information); 3 Verified.

Fake Lab case ratings

CASE	RATING	WHY
1 The giant pumpkin	Verified	First posted by the Fair's official account; the newspaper confirmed with a quote from the judge; many attendees posted.
2 The flooded hallway	Real but miscaptioned	An image search shows the same picture from three years ago in another town, and the school's official alert system says nothing.
3 The principal's voice message	False (possible voice imitation or scam)	The school's site and alert system do not mention a closure, the number is unknown, and the office did not send it. Check with the office; do not reply with personal information.
4 The star skater's offer	False (a fake endorsement)	His official account has no such video and a fact-check card calls it fake. Even if the video looks perfect, check official channels.
5 The panic call	False (a voice imitation scam)	A code word was requested and not given, and a call to the known number reached the relative safe at home. The code word and callback worked.
6 The unexpected owl	Verified	Three local news sites, a wildlife agency post and several photographers from different angles. A reminder not to dismiss real things just because they are surprising.

After each rotation ask: "Which check gave you the answer?"

Caption Flip prompt answer: who took it, when and where.

Do not use AI detector tools on students' work to accuse them; they are unreliable.